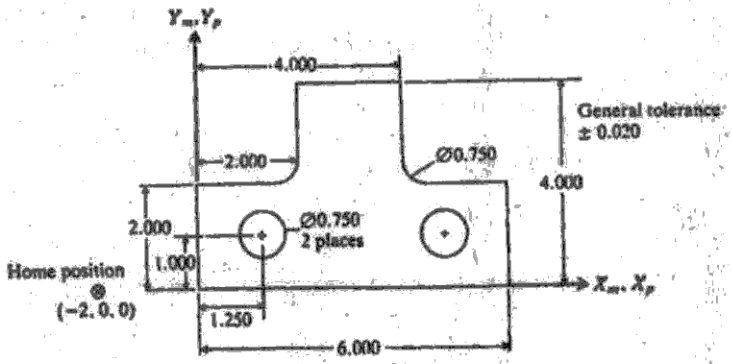

FIGURE-2

(b) What is NC programming ? Write about NC programming languages and applications. 10

(c) Explain the necessity of G-codes and M-codes in CAM, giving a suitable example. What is meant by Datum Setting ? 10

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ME (2nd Semester)

Examination, April-May, 2012

Branch : Mechanical

Specialization : Design

CAD/CAM APPLICATION

Time Allowed : Three Hours

Maximum Marks : 100

Minimum Pass Marks : 40

lote : Attempt all questions. Each question carries equal marks. Assume suitable data and provide self-explanatory figures wherever required.

. 1. Answer any two :

(a) Enlist the merits & demerits of the various interactive input devices used in CAD/CAM hardware. Explain them in brief. 10

(b) What are the types of graphics displays in CAD/CAM system ? Compare their capabilities, advantages & disadvantages. 10

(c) Discuss the CAD/CAM systems to engineering design considering the following issues : 10

(i) Cost : capital investment, running cost & maintenance cost

(ii) Capabilities : geometric modeling and programming

(iii) Convenience of use

(iv) Future improvements

Q. 2. Attempt any two :

(a) What are the various software modules in CAD/CAM ? Discuss and compare them in brief. 10

(b) Find the equation of an open quadratic B-spline curve defined by five control points. 10

(c) A cubic Bezier curve is to be divided by a designer into two segments. Find the modified polygon points for each segment. 10

Q. 3. Attempt any two :

(a) What is parametric representation of analytic surfaces ? Find the minimum distance between a point in space and a plane surface. 10

(b) Give a short note on Hermite Bicubic Surface. Also explain the necessity of parametric representation of synthetic surfaces in engineering design. 10

(c) Find the equivalent bicubic formulation of a cubic Bezier surface patch. 10

Q. 4. Solve any two :

(a) Prove that if a point to be rotated about a given axis of rotation lies on the axis, the point does not change position in space and, therefore, its coordinates do not change. 10

(b) For a geometric model shown in figure-1, a user defines the XYZ coordinate system shown as the MCS of the model database. The user later defines the two WCSs {w} and $\{w_1\}$ shown to be able to construct the two circles whose centres are C and C_1. Both C and C_1 are the centre points of the respective faces of the model. Determine : 10

(i) The mapping matrices $[T_w]$ and $[T_{w_1}]$ that map points from any one of the two WCSs to the MCS.

(ii) The coordinates of the centres C & C_1 as they are stored in the model database.

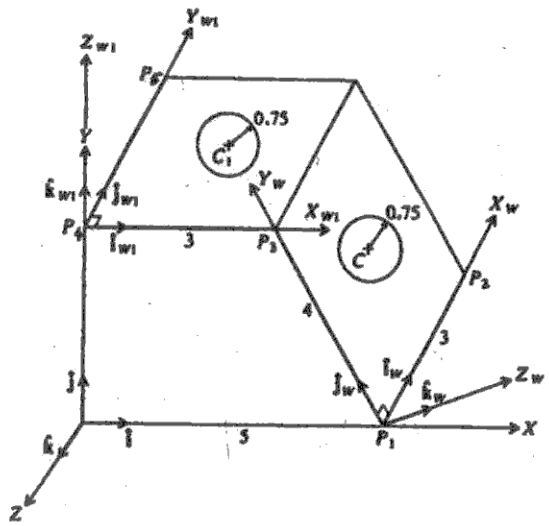

FIGURE-1

(c) How can a Bezier curve, B-spline curve, Hermite bicubic surface, Bezier surface, and B-spline surface be scaled uniformly ? **10**

Q. 5. Answer any two :

(a) Write an APT program to mill the contour of the part as shown in figure-2. Use an end-mill cutter with a diameter of 0.75 inch. Assuming the part is made of steel. Cutting speed of the cutter is 580 rev./min & feed is 2.3 inch/min. Name of milling machine is MILL 5. **10**